

7 Three planes have equations

$$x + 2y - 3z = 0,$$

$$-x + 3y - 2z = 0,$$

$$x - 2y + kz = k,$$

where k is a constant.

(a) For the case $k = 0$, the origin lies on all three planes.

Use a determinant to explain whether there are any other points that lie on all three planes in this case. **[2]**

(b) You are now given that $k = 1$.

(i) Show that there are no points that lie on all three planes. **[3]**

(ii) Describe the geometrical arrangement of the three planes. **[1]**